

YA-NING (YANNI) WU

PhD Candidate in Cognitive Sciences · University of California, Irvine

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RESEARCH PROFILE

I am a computational cognitive scientist studying how biological and artificial agents infer temporal structure, form predictions, and synchronize action. My research combines computational modeling with behavioral experimentation to investigate rhythm perception and temporal prediction, with complementary work in speech timing and auditory neuroscience. I develop and empirically compare mechanistic models of temporal tracking and build open-source experimental tools for simulated agents.

EDUCATION

University of California, Irvine

Ph.D. in Cognitive Sciences

Advisor: Gregory Hickok, Auditory & Language Neuroscience Lab

GPA: 3.97/4.00

Irvine, CA

Expected 2027 | Sep 2022 - Present

University of California, San Diego

B.S. in Cognitive & Behavioral Neuroscience, Magna Cum Laude

GPA: 3.93/4.00 · Department Honors · Provost's Honors

Honors Thesis: Sound-Symbolic Effects of Labels on Size Perception Memory

La Jolla, CA

Sep 2016 - Jun 2019

RESEARCH EXPERIENCE

Auditory & Language Neuroscience Lab · University of California, Irvine

Graduate Researcher · Advisor: Gregory Hickok

Sep 2022 - Present

Computational models of rhythm and sensorimotor synchronization

- Develop and compare computational models of rhythm tracking, temporal prediction, and sensorimotor synchronization.
- Design behavioral paradigms to investigate rhythm perception, temporal prediction, and motor synchronization.
- Implement predictive and reactive synchronization models in embodied e-puck2 robots.
- Developed SYNCOFLY, a browser-based multi-agent platform for studying collective rhythm synchronization, in collaboration with Dr. Jeffrey Krichmar's Cognitive Ant eater Robotics Laboratory.

Speech rhythm and temporal organization

- Investigated English lexical stress production in native American English and L1-Mandarin speakers, analyzing cross-linguistic differences in acoustic cue weighting.
- Studied the effects of delayed auditory feedback on segmental and prosodic representations in speech production.

Brain and Language Laboratory · National Central University

Research Assistant

Aug 2019 - Mar 2022

- Conducted MEG/EEG analyses of auditory, speech, and language processing using MNE-Python and MATLAB.
- Developed neural decoding pipelines combining machine learning with nonlinear signal decomposition to identify neural patterns in auditory and speech signals.
- Studied speech perception, lexical processing, and auditory signal denoising.

Center for Research in Language · University of California, San Diego

Research Assistant

Aug 2017 - Jun 2019

- Conducted an honors thesis investigating sound symbolism and memory for referent size, combining behavioral experimentation with computational analysis.
- Developed analysis pipelines for research on language processing and auditory perception.
- Contributed to research on morphological complexity using artificial-language paradigms.

Neuroanatomy Laboratory · National Taiwan University

Summer Research Intern

Jun 2018 - Sep 2018

- Assisted with experimental research on neural development and subplate neuron organization using histological staining and imaging techniques.

George Laboratory · The Scripps Research Institute

Research Assistant

Apr 2018 - Dec 2018

- Supported experimental research on addiction and treatment in animal models.

PUBLICATIONS

Wu, Y., Hickok, G., & Krichmar, J. L. (2026). SYNCOFLY: An Interactive Platform for Collective Rhythm Synchronization. *Simulation of Adaptive Behavior (SAB 2026)*.

Hsu, C.-H., & Wu, Y.-N. (2021). Application of Empirical Mode Decomposition for Decoding Perception of Faces Using Magnetoencephalography. *Sensors*, 21, 6235.

Hsu, C.-H., Wu, Y.-N., & Lee, C.-Y. (2021). Effects of Phonological Consistency and Semantic Radical Combinability on N170 and P200 in the Reading of Chinese Phonograms. *Frontiers in Psychology*.

SELECTED CONFERENCE PRESENTATIONS

Wu, Y. -N., Hickok, G., & Krichmar, J. L. (2026). SYNCOFLY: An Interactive Platform for Collective Rhythm Synchronization. *Simulation of Adaptive Behavior (SAB 2026)*.

Wu, Y. -N. (2026). When Perception Outperforms Action: Dissociating Rhythm Perception from Motor Synchronization. Talk presented at the Society for Music Perception and Cognition (SMPC 2026).

Wu, Y. -N. & Hickok, G. (2025). Self-Monitoring through Rhythm: Probing Prosodic Planning with Delayed Auditory Feedback. Poster presented at the Society for the Neurobiology of Language (SNL 2025).

Wu, Y. -N., Chang, C., & Hickok, G. (2024). Exploring Cross-Categorical Pitch Shift Effects on Mandarin Tone Production. Poster presented at the Society for the Neurobiology of Language (SNL 2024).

Chen, P.-H., Wu, Y.-N., Hsu, C.-H., & Lee, C.-L. (2021). Resolving Lexical Ambiguity Enhances Frontal Positivity: Data from Event-Related Potentials and Magnetoencephalogram. Poster presented at the Cognitive Neuroscience Society Annual Meeting.

Hsu, C.-H., Wu, Y.-N., & Yen, T. (2020). MEG Response of Left Temporal Cortex to Modifier–Head Compounds. Annual Meeting of the Taiwan Society of Cognitive Neuroscience.

Wu, Y. -N. & Semenuks, A. (2020). Sound-Symbolic Effect of Labels on Size Perception Memory. Proceedings of the 13th International Conference on the Evolution of Language (EvoLang XIII).

RESEARCH SOFTWARE

SYNCOFLY · *Collective Rhythm Synchronization*

GitHub: github.com/yanni-yanni/syncofly

Open-source browser-based platform for studying collective temporal coordination in multi-agent systems. Supports predictive and reactive agents, multiple perturbation parameters, and real-time behavioral logging.

COMPUTATIONAL & EXPERIMENTAL SKILLS

Programming: Python (NumPy, SciPy, Pandas), R, MATLAB, JavaScript/HTML/CSS

Audio & signal processing: Praat · Audapter · Web Audio API

Neuroimaging: MEG/EEG · MNE-Python · MATLAB · FreeSurfer · SPM12

Robotics & embodied systems: e-puck2

LANGUAGES

English · Mandarin Chinese · Japanese · German